Data Structures and Algorithms Lab
Assignment 2

Experiment 2: To experiment with the concept of pointers, structure, and dynamic memory allocation to realize linked lists, their types, and application.

Instruction: Make a pdf file of the program code in C and its output.

Name:
SAP ID:
Batch:
Link of your GitHub Account:
Course: B.Tech. CSE (Sem II)
Subject: DSA Lab
Date: 17 February 2026
Note: Do all the implementations in C programming.

1. Write a C program to traverse a **singly linked list** and count the number of nodes present in the list.

2. Write a C program to implement a **singly linked list** and search for a given element in the list. If the element is found, display the position of the element in the linked list; otherwise, display element does not exit.

3. Write a C program to implement **insertion** operations in a **singly linked list**. The program should perform the following operations:
   - Insert a new element at the beginning of the linked list.
   - Insert a new element at the end of the linked list.
   - Insert a new element after a given node in the linked list.

4. Write a C program to implement **deletion** operations on a **singly linked list**. The program should support deleting an element from:
   - The beginning of the linked list
   - The end of the linked list
   - After a given node (specified by its value)

5. Write a C program to reverse the data of the nodes of a singly linked list without changing the links between the nodes.

6. Write a C program to implement a **Circular Singly Linked List** and perform all basic operations on it. The program must support the following operations:
   - Creation of a Circular Linked List
   - Insertion of a node (At the beginning and at the end of the list)
   - Deletion of a node (From the beginning and from the end of the list)
   - Traversal and display of the circular linked list and counting the number of nodes in the list.

7. Write a C program to implement a **Doubly Linked List**. The program should perform all basic operations on the list. The program must support the following operations:
    - Creation of a Doubly Linked List, display and counting the number of nodes
    - Insertion of a node (At the beginning, At the end, After a given node)
    - Deletion of a node (From the beginning, From the end, of a given node)
    - Traversal of the list (Forward traversal and Backward traversal)

8. Write a C program to implement a **Doubly Circular Linked List**. The program should perform all basic operations on the list. The program must support the following operations:
    - Creation of a Doubly Circular Linked List, display and counting the number of nodes
    - Insertion of a node (At the beginning, At the end)
    - Deletion of a node (From the beginning, From the end)

9. Write a C program to represent and evaluate a Polynomial using a Linked List. Each node should represent a term containing coefficient and exponent.

10. Write a C program to maintain Student Records using a Linked List. Each node should store student details such as roll number, name, and marks. Implement operations to insert, delete, search, and display records.